TECHNICAL PROPOSAL: QUANTUM-ECCENTRIC LAGRANGIAN DYNAMICS IN DITRITIUM-BASED IQ-9 HIGGS PROCESSORS FOR RESILIENT DATA ASSIMILATION IN EXTREME HELIOPHYSICS ENVIRONMENTS

Solicitation: NNH25ZDA001N-RADIANT (B.5)

Document Type: Scientific/Technical/Management Section

Status: Anonymized for Dual-Anonymous Peer Review (DAPR)

1. EXECUTIVE SUMMARY

This proposal introduces the **IQ-9 Higgs Architecture**, a revolutionary computing paradigm designed for Heliophysics missions operating in extreme radiation environments. By integrating a superheavy Ditritium substrate ($Z = 155$) with non-linear Lagrangian dynamics, the proposed technology provides a deterministic, fault-tolerant solution for real-time data assimilation. The architecture leverages Google Qualtran for resource estimation and IBM Qiskit/Google Cirq for trajectory validation, ensuring computational sovereignty and data integrity during high-energy solar particle events (SPE).

2. SCIENTIFIC MERIT AND HELIOPHYSICS RELEVANCE

2.1 The Challenge of Solar Radiation

Heliophysics instruments in proximity to the Sun (e.g., Parker Solar Probe, Solar Orbiter) face severe hardware degradation and telemetry corruption. Traditional silicon-based logic suffers from Single Event Upsets (SEUs) and total ionizing dose (TID) failures.

2.2 The Ditritium Solution ($Z = 155$)

The proposed IQ-9 architecture replaces traditional semiconductors with a high-tenacity Ditritium matrix. The stability of this substrate is governed by the **Diadema Tenacity Matrix**, ensuring that thermal and nuclear stress remain within operational limits:

$$\frac{d\sigma_{th}}{dT} = E \cdot \alpha$$

Where σ_{th} is the thermal stress and α is the expansion coefficient of the Z=155 matrix.

2.3 Lagrangian Reality Anchoring

To prevent computational "hallucinations" and ensure deterministic outcomes, the IQ-9 core implements a 1-jet prolongation $J^1\pi$ on the manifold Z:

$$\dim Z = n + 1 + m + (n+1)m$$

This mathematical grounding allows for the tracking of quantum trajectories with a fixed "Reality Anchor" at $i = 1, q = 1$, neutralizing the effects of external electromagnetic interference.

3. TECHNICAL APPROACH AND ARCHITECTURE

3.1 N-Local Circuit Mesh

The IQ-9 Processor utilizes an **N-local circuit mesh** that alternates single-qubit rotation layers (RX) with multi-qubit entanglement layers (CZ). The performance is measured by the **Ditritium Energy Index** (E_{Dt}):

$$E_{Dt} = \frac{\mathcal{O} \cdot \mathcal{D}}{n \cdot \zeta}$$

Where $\mathcal{O}$ is total gate count, $\mathcal{D}$ is depth, and $\zeta = 1.55$.

3.2 Integrated Software Stack

The architecture is supported by a robust Python-based ecosystem that integrates:

- **Qualtran (Google):** For Bloq-based resource estimation and Galois Field ($GF(p^m)$) arithmetic.
- **Cirq (Google):** For physical trajectory mapping.
- **Qiskit (IBM):** For logical gate transpilation and error correction.

4. INTEGRATED ALGORITHMS AND SOURCE CODE

This section provides the complete algorithmic implementation of the IQ-9 Higgs Core system, as validated in laboratory environments.

4.1 IQ-9 Core System (Native OS)

The following code manages the nuclear allocation and the **Prime Mesh** status of the processor.